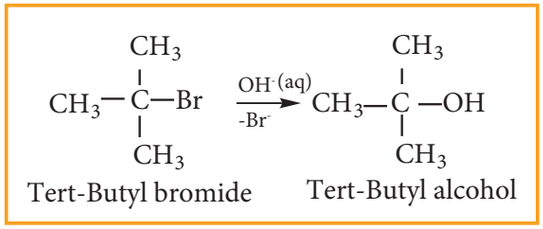
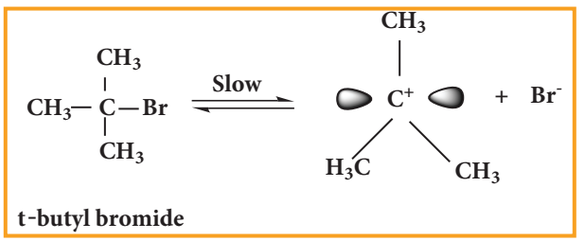
We understand S_N1 reaction mechanism by taking a reaction between tertiary butyl bromide with aqueous KOH.



This reaction takes place in two steps as shown below

Step - 1 Formation of carbocation

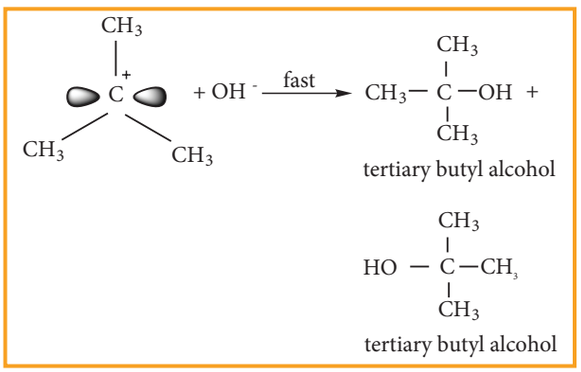
The polar C - Br bond breaks forming a carbocation and bromide ion. This step is slow and hence it is the rate determining step.



The carbocation has 2 equivalent lobes of the vacant 2p orbital, so it can react equally rapidly from either face

Step - 2

The nucleophile immediately reacts with the carbocation. This step is fast and hence does not affect the rate of the reactions.



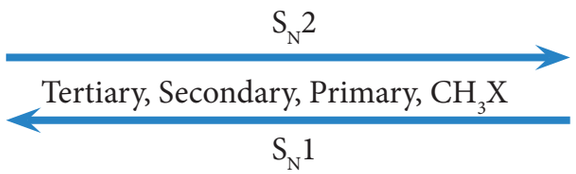
As shown above, the nucleophilic reagent OH^- can attack carbocation from both the sides.

In the above example the substrate tert-butyl bromide is not optically active, hence the obtained product is optically inactive. If halo alkane substrate is optically active then, the product obtained will be optically inactive racemic mixture. As nucleophilic reagent OH^- can attack carbocation from both the sides, to form equal proportion of dextro and levorotatory optically active isomers which results in optically inactive racemic mixture.

Example

Hydrolysis of optically active 2 - bromo butane gives racemic mixture of ± butan-2-ol

The order of reactivity of haloalkanes towards S_N1 and S_N2 reaction is given below.



Evaluate Yourself

4) neo-pentyl bromide undergoes nucleophilic substitution reactions very slowly - justify.

2) Elimination reactions

When a haloalkane containing a hydrogen on β carbon is treated with an ethanolic solution of potassium hydroxide, an alkene is formed. In this reaction a double bond between α and β carbon is formed by

Unit 14.indd 236 23-12-2021 17:22:50